IS492 Spring 2026
CheckPoint 4

Adaptfit

An injury aware workout adapter and generator that helps users modify training plans and avoid secondary injuries

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USE ON YOUR PHONES NOW!!

Demo: https://deploy-app-sigma.vercel.app

GitHub repo link:
https://github.com/IS492-SP26/team-project-injury-aware-workout-planner

Our Agenda

- Recap & Final Goals
- Evaluation Design Overview
- Study Materials & Protocol
- Quantitative Results
- Qualitative Insights
- Interpretation & Discussion
- Limitations, Risks & Ethics
- Conclusion & Future Work

Recap & Final Goals

Target Users

Regular gym users with non-severe injuries (e.g., knee or shoulder) and in the recovery phase, but not in formal rehab, and want to stay active safely.

Problem Statement

Current fitness apps don’t adapt to users with injuries. We address the lack of safe, personalized workout guidance for these users.

User Feedback Highlight

Task completion rate is 100%.
Error rate < 10%.
Users rate 4.40/5.0 for the injury assessment questions were highly rated for their relevance

Evaluation Design Overview

- **Goal:** Assess whether AdaptFit AI is understandable, useful, and trustworthy for injury-aware workout modification.

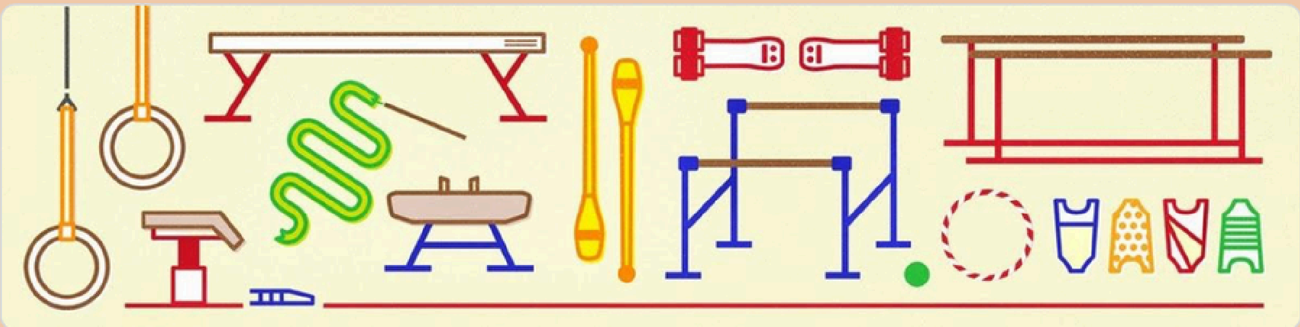
- **Participants**:
  - 21 ppl
  - Mixed user backgrounds: general gym users, beginners, athletes/dancers, and users with injury-related concerns

- **Study tasks- Participants reviewed the AdaptFit AI workflow and evaluated whether it helped them:**
  - Use the app to finish at least 1 workout video
  - Understand the app's safety role
  - Enter injury/readiness information
  - Interpret risk labels and substitutions
  - Decide how to modify workouts safely

- **Metrics collected:**
  - Task success rate, error rate
  - Time-on-task
  - System Usability Scale - 5 score Lite
  - Satisfaction, usefulness
  - Qualitative feedback themes

Study Materials & Protocol

Task prompts



AdaptFit AI Post-Task Evaluation Survey

Consent Statement

Thank you for participating in our study. We are testing **AdaptFit AI**, an injury-aware workout planning prototype for a class project.

This tool is designed to support general workout planning and modification for non-severe injuries. It is **not a medical tool** and does **not replace professional medical advice, diagnosis, treatment, physical therapy, or emergency care**.

During this study, you will be asked to complete a short task using the prototype and answer a few survey questions about your experience. Your feedback will help us evaluate the system's usability, safety clarity, usefulness, and trustworthiness.

Your participation is voluntary. You may skip any question or stop participating at any time. Your responses will be used only for this course project and will be anonymized in our final report.

*** By continuing, you confirm that you understand the purpose of this study and agree to participate.***
The survey takes approximately 5 mins. Thank you for your time! ❤️

Are you a medical professional? If yes - please add more details about your level of education and practice. *

Your answer

Are you a frequent gym-goer? If yes - please add more description about how frequently you visit the gym and how long have you been going to the gym now. *

Your answer

Please rate each statement using the following scale:

1. I clearly understood that AdaptFit AI is a support tool and not a replacement for professional medical advice. *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	◉	Strongly agree

2. The app made it clear when I should pause and consult a professional instead of continuing workout modifications. *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

Survey Interface - Google Form

Section 2 of 2

Optional Follow-Up Questions

Description (optional)

1. What was the single most helpful part of the experience?

Long answer text

2. What part of the experience felt least reliable or most risky?

Long answer text

3. What should be improved first to make the system safer and more trustworthy?

Long answer text

Thank you for your time and feedback. Your responses will help us better understand what works well, what feels unclear or risky, and how AdaptFit AI can be improved in future iterations.

Description (optional)

Quantitative Results

Task Completion Rate

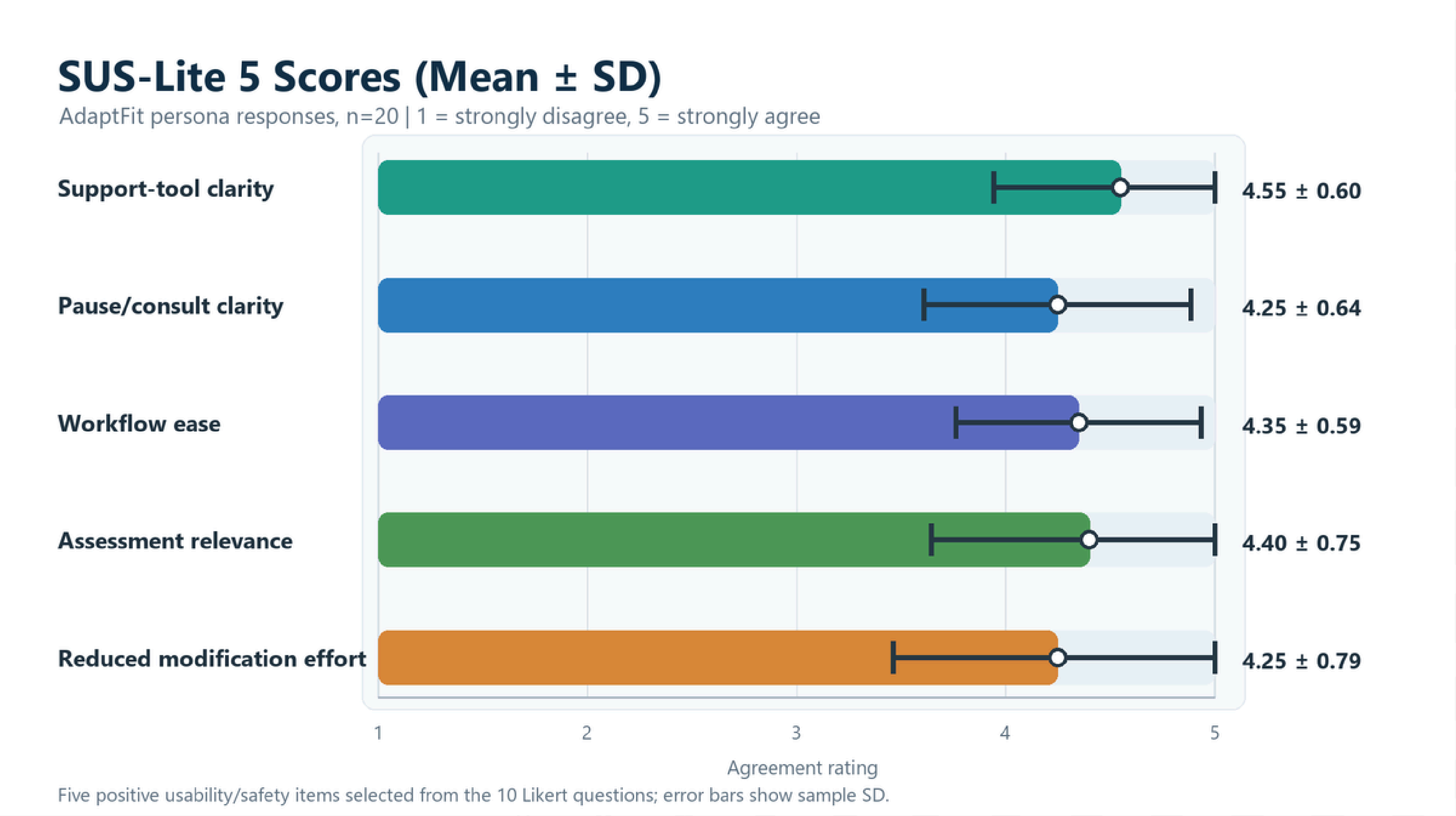
100% [All users completed]

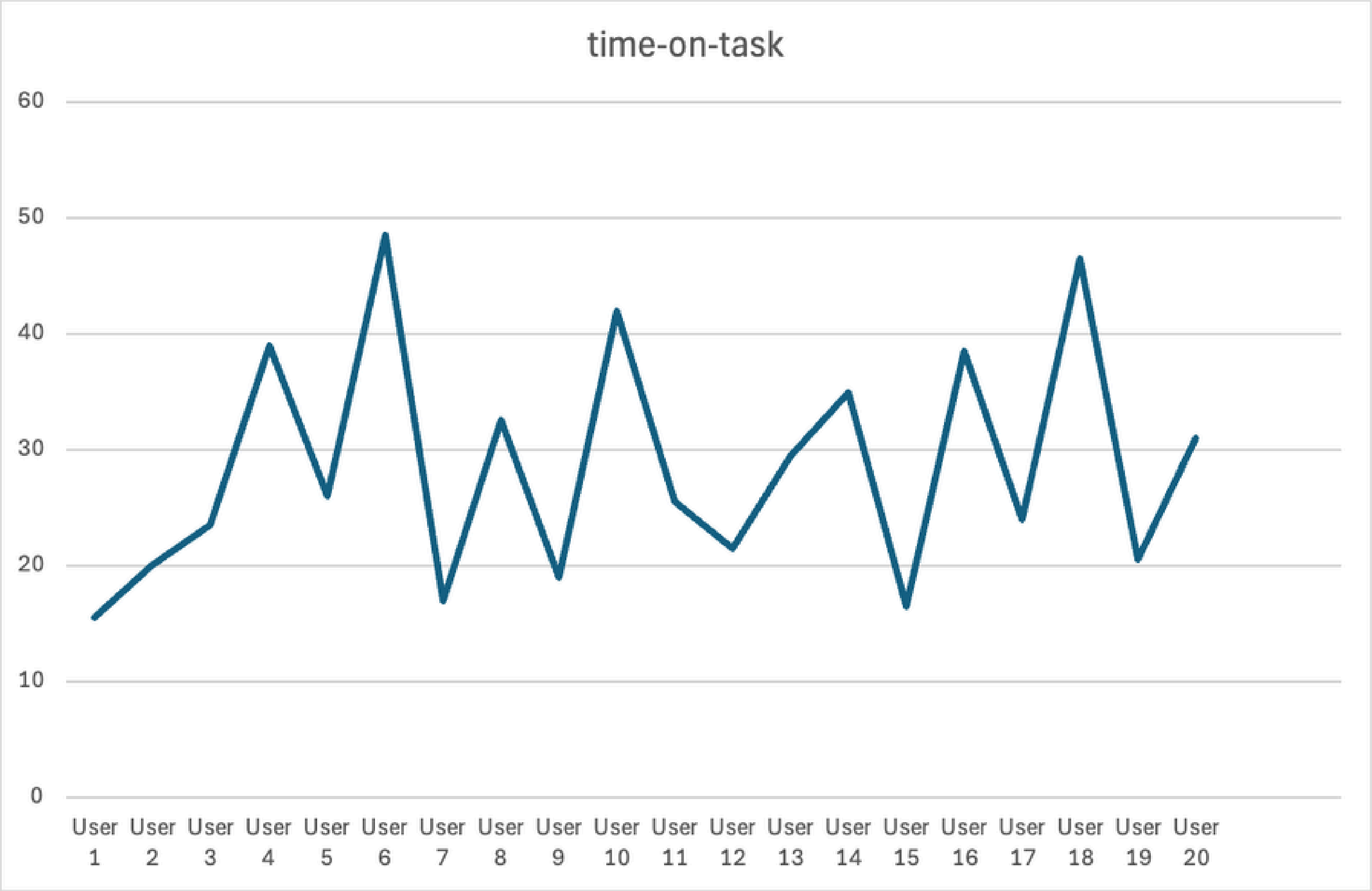
Time-on-task comparisons

Max ~ 48min/Min ~ 15min/Avg ~ 28min

Error Rate

9.5% => **2 out of 21** users experienced a timestamp mismatch





Qualitative Insights

Positive themes

- Readiness scores and risk drivers
- Useful workout flow grouping
- Relevant injury assessment

- “The readiness score and risk drivers were most helpful.”
- “The workout flow panel was useful because it grouped exercises by time window ”

Failures

- Video timestamp mismatch for longer videos
- Limited injury profile depth

- “There is video timestamp issue. The last three timestamps points to the wrong exercise”
- “For lower back tightness, I wanted more specific screening questions about bending.”

Limitations, Risks & Ethics

Limitations

- We have only 21 responses.
- The app supports workout adaptation, but does not diagnose injuries or assess exercise form.
- Risk scoring depends on self-reported pain, movement limits, and recovery details, which may be incomplete or inaccurate.
- Some injury contexts, especially sport-specific return-to-play, need more detailed screening.

Risks

- Users may over-trust AI suggestions and treat them like medical advice.
- Incorrect or overly broad substitutions could still aggravate an injury.
- Missing red flags could delay professional care.
- Video timestamp mismatch could cause users to modify the wrong exercise.

Ethical Safeguards

- Clearly position AdaptFit as a support tool, not a replacement for clinicians.
- Use conservative recommendations when pain, recent injury, or uncertainty is high.
- Add strong pause/consult prompts for severe, worsening, or unclear symptoms.
- Prioritize transparency: explain why an exercise is risky and why an alternative is safer.

Conclusion & Future Work

MAIN TAKEAWAYS -

- AdaptFit improved clarity and effort: users could follow and the adapted plan reduced the work of deciding what to skip/modify.
- Users felt safer when the system clearly stated injury readiness, including when to pause and consult a professional.

CONTRIBUTION TO HUMAN-AI COLLAB

- Shared decision-making model: the AI proposes injury-aware modifications (risk labels + safer substitutes), while the user/clinician retains control—AI as support, not authority.
- Calibrated trust: clear boundaries (not medical advice), “not cleared” outcomes, and explanations help users appropriately rely on the tool without over-trusting it.

FUTURE IMPROVEMENT

- Improve “not cleared” flow with next steps (what to do now, what info is missing).
- Better comparability view (original vs adapted) and quick edit/override controls.
- Expand dataset across injury types, recovery stages, and user experience levels.
- Validate substitutions with expert review and (where possible) outcomes tracking.

Cursor, VsCode, Claude, ChatGPT, Gemini, Stitch, Figma, Xcode, Google Forms, Google Sheets, Google Colab, GitHub, Vercel, WhatsApp

Acknowledgments & Contributions

Emma
(Product Management & Evaluation)
Responsibilities: Product framing, safety positioning, survey design, results analysis, insight synthesis

Prisha
(UX& Product Development)
Responsibilities: Built the User interface and connected that to the backend. Maintained the code throughout the process. Built xcode project for live app view in the initial phases.

Ocean
(Software & Prompt Development)
Responsibilities: Optimized the system prompt architecture, refined workout adjustment scripts and input data structures.

Vinit
(Software Development [Backend & AI])
Responsibilities: Built the API flow from the Supabase backend. Built the prompts for video timestamping and yt in-app player with dynamic suggestions.

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Thank You